

LAB EXPERIMENTS

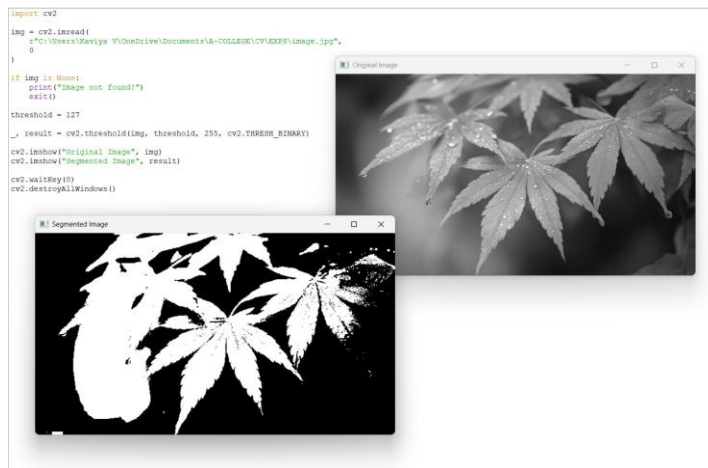
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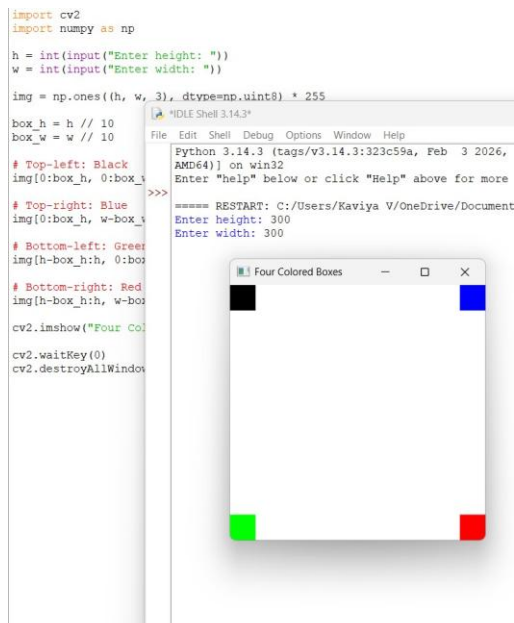
COURSE CODE: ITA0512

COURSE NAME: COMPUTER VISION

31. Implement a Segmentation algorithm using Open CV to segment the given input image based on the given threshold values.



32. Write a Python function to create a white image size entered by the user and then create 4 boxes of Black, Blue, Green and Red respectively on each corner of the image. The size of the colored boxes should be 1/10th the size of the image. (HINT: the arrays of ones and zeros can be in more than 2 dimensions).



33. Write a Python function to create a white image size entered by the user and then create a shape of Rectangle using Open CV.

```
import cv2
import numpy as np

h = int(input("Enter height: "))
w = int(input("Enter width: "))

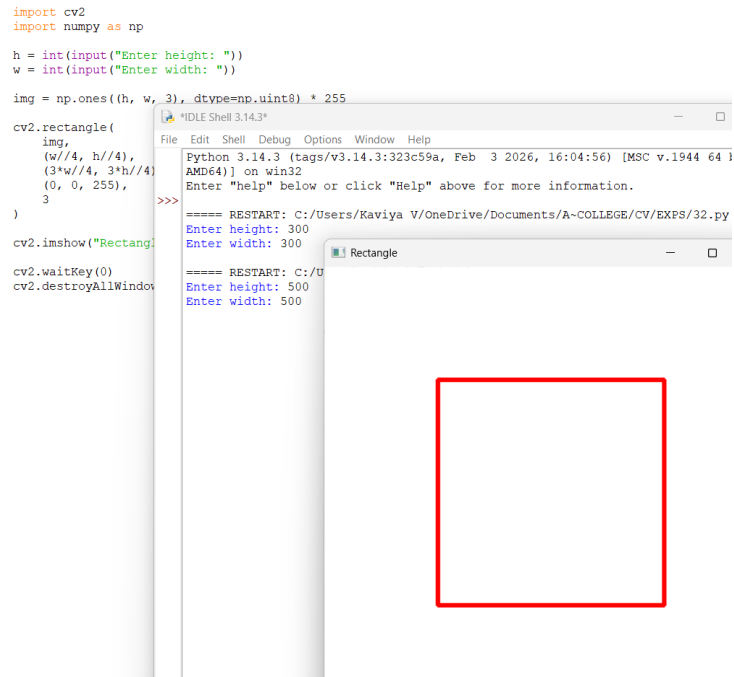
img = np.ones((h, w, 3), dtype=np.uint8) * 255

cv2.rectangle(
    img,
    (w//4, h//4),
    (3*w//4, 3*h//4),
    (0, 0, 255),
    3
)

cv2.imshow("Rectangle", img)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

(Note: The original image shows a red rectangle, but the code specifies a blue rectangle. The output image below shows a red rectangle, suggesting the code in the image might have been misread or the image was edited.)



34. Write a Python function to create a white image size entered by the user and then create a shape of Circle using Open CV.

```
import cv2
import numpy as np

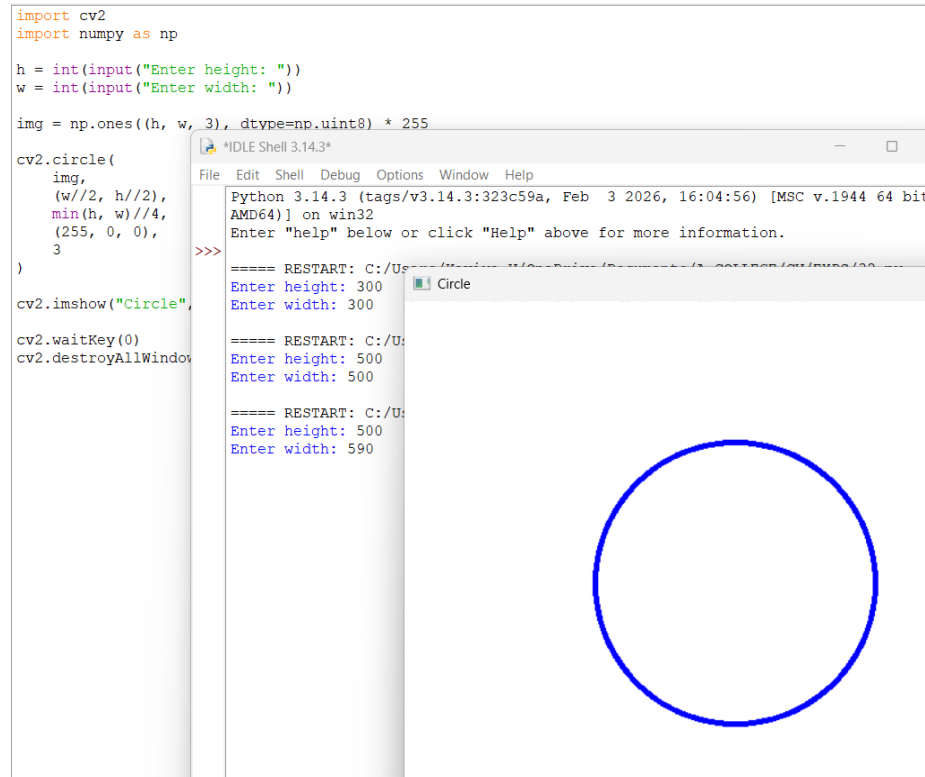
h = int(input("Enter height: "))
w = int(input("Enter width: "))

img = np.ones((h, w, 3), dtype=np.uint8) * 255

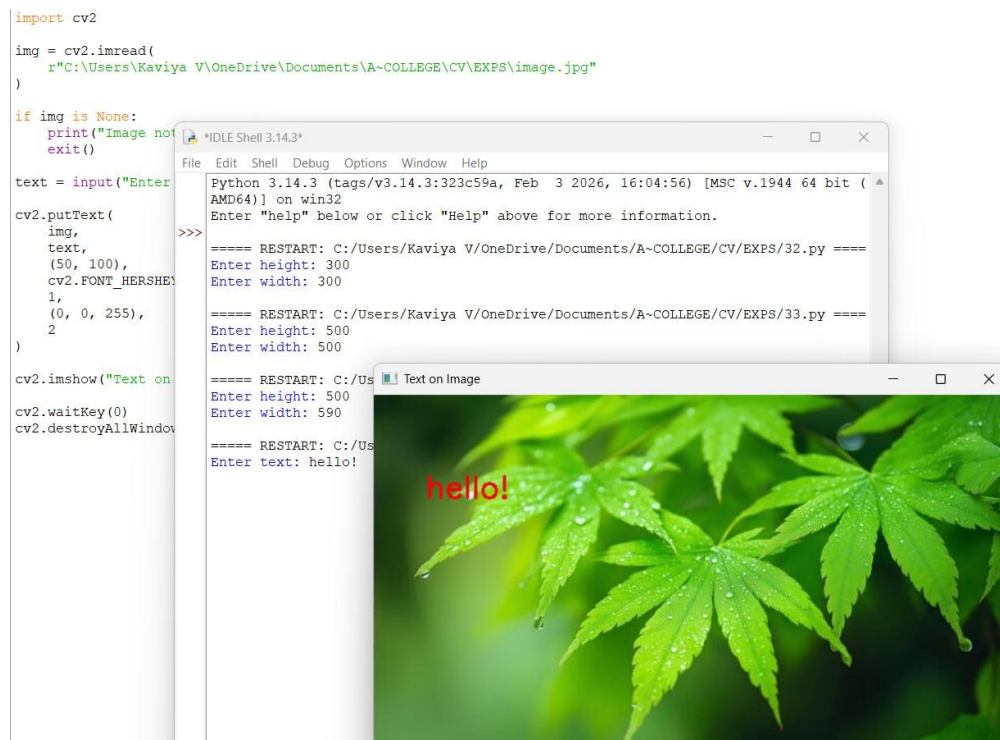
cv2.circle(
    img,
    (w//2, h//2),
    min(h, w)//4,
    (255, 0, 0),
    3
)

cv2.imshow("Circle", img)

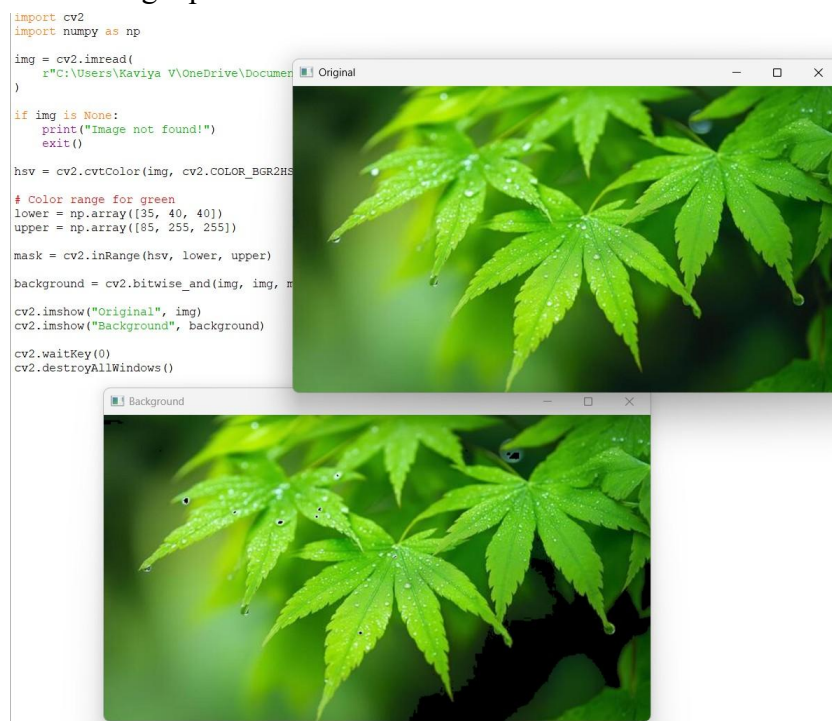
cv2.waitKey(0)
cv2.destroyAllWindows()
```



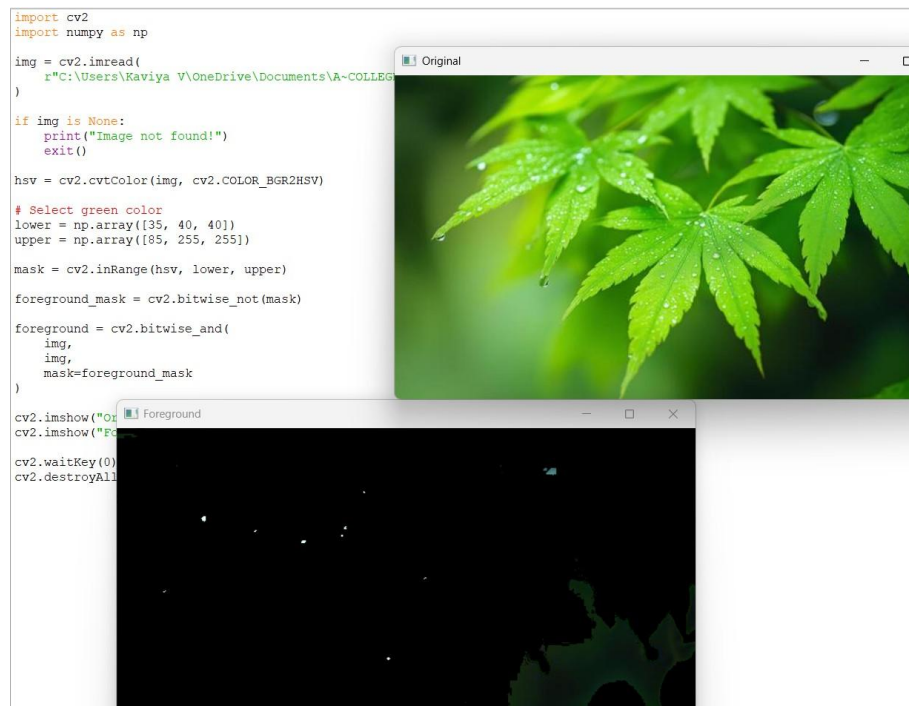
35. Write a Python function to create a text string entered by the user that must be appeared on the given image using Open CV.



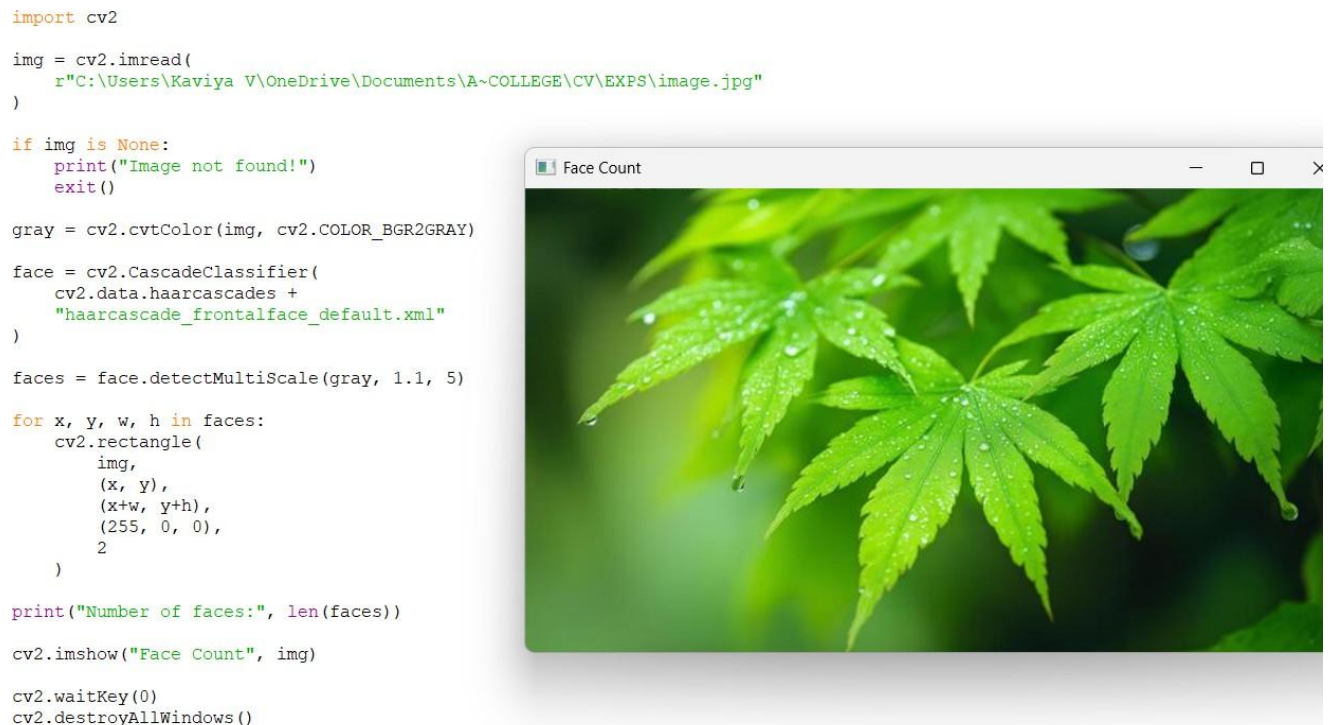
36. Write a Python function to subtract the background of the given input image based on color levels using Open CV.



37. Write a Python function to subtract the foreground of the given input image based on color levels using Open CV.



38. Write a Python function to Count the number of faces for the given input image using Open CV.



39. Write a Python function to play the given video in reverse mode in slow motion.

```
import cv2

cap = cv2.VideoCapture(
    r"C:\Users\Kaviya V\Downloads\video.webm"
)

if not cap.isOpened():
    print("Video not found!")
    exit()

frames = []

while True:
    ret, frame = cap.read()

    if not ret:
        break

    frames.append(frame)

cap.release()

# Play frames in reverse
for frame in reversed(frames):

    cv2.imshow("Reverse Slow Motion", frame)

    # Higher waitKey value = slower video
    if cv2.waitKey(100) & 0xFF == ord('q'):
        break

cv2.destroyAllWindows()
```



40. Write a Python function to extract the text from videos.

```
import cv2
import pytesseract

cap = cv2.VideoCapture(
    r"C:\Users\Kaviya V\Downloads\video.webm"
)

if not cap.isOpened():
    print("Video not found!")
    exit()

# Read one frame
ret, frame = cap.read()

if ret:
    text = pytesseract.image_to_string(frame)

    print("Extracted Text:")
    print(text)

    cv2.imshow("Video Frame", frame)
    cv2.waitKey(0)

cap.release()
cv2.destroyAllWindows()
```